

From *In Silico* to Bench: Deep Learning-Based *De Novo* Design, Computational Validation, and Synthesis of CD109-CD26-Targeting Peptides for T-Cell Modulation

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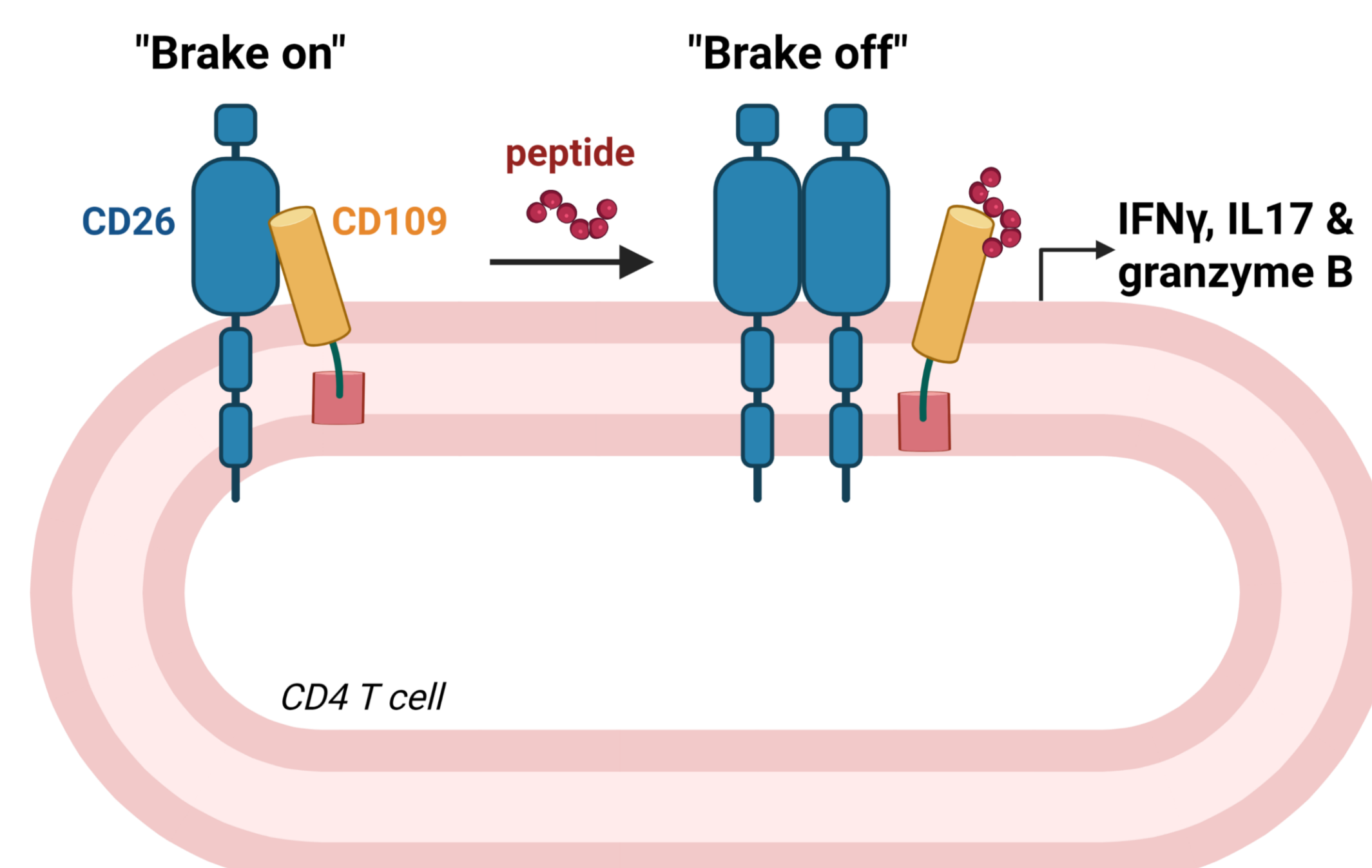
COLLABORATE. INNOVATE. ACCELERATE.

Abstract

CD26, a non-canonical co-stimulator, drives polyfunctional Th1/Th17 responses, and CD109 acts as an intrinsic brake by directly binding to CD26. Disrupting the CD109-CD26 interface releases this brake and enhances effector function, boosting CAR T-cell anti-tumor profile. A CD26-derived decoy peptide offered proof of concept. To move beyond this lead peptide, we built a multi-layer computational pipeline combining deep learning-based *de novo* design, biophysical filters, and docking, narrowing 5000 candidates to 13 peptides for synthesis. Peptides were synthesized on an automated CEM Liberty Blue SPPS. Future work will evaluate their ADME properties and ability to block the CD109-CD26 interaction in cell-based assays.

Introduction & Rationale

- CD4 T-cell activation is regulated by a balance of co-stimulatory and inhibitory signals
- CD26 is a non-canonical co-stimulator; on dimerization it drives polyfunctional Th1/Th17 cells producing IFN γ , IL-17A, GM-CSF, and granzyme B and in anti-tumor immunity
- CD109 acts as an activation-induced intrinsic brake that directly binds to CD26 and dampens its co-stimulatory output
- Disrupting the CD109-CD26 interface releases this brake, enhancing effector function and improving CAR T killing of tumors



Lead peptide showed binding promise but was limited by poor stability.

Aim: Apply David Baker's *de novo* protein design (ProteinMPNN) and other computational tools to design stable, low-immunogenicity, high-affinity CD109 binder peptides to use for tuning T-cell immunity.

Computational Pipeline

Lead peptide (*HIYTHMSHFQKCFSLP*): -41.12 kcal/mol docking, GRAVY 0.035, 9 high-risk peptidase sites.

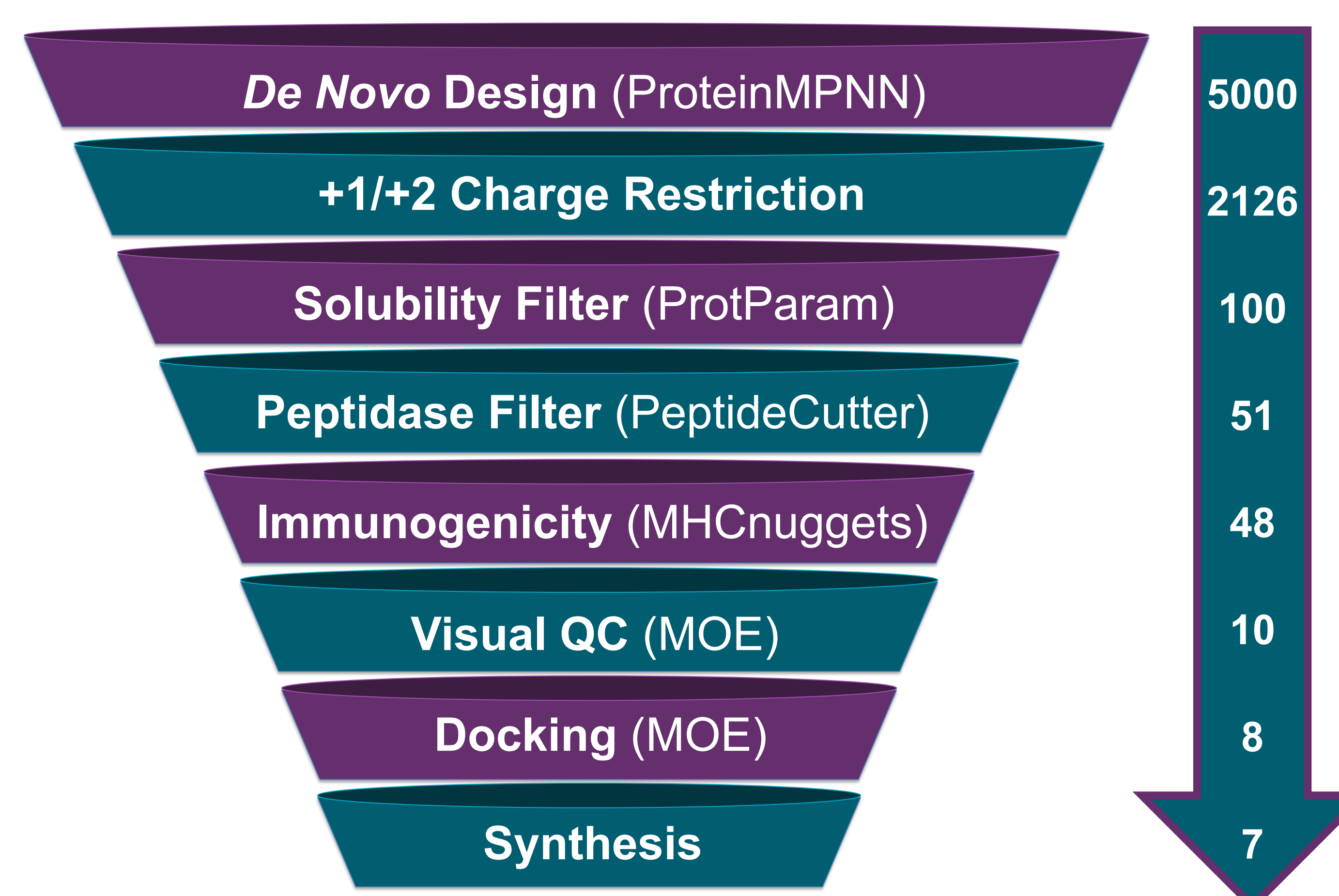
De Novo Design: 5,000 analogs generated via ProteinMPNN. CD109 anchors fixed; net +1/+2 charge enforced; bulky/acidic residues penalized.

Docking & Final Validation: MOE visual QC removed broken backbones. Rigid-body protein-protein docking eliminated high-energy conformations.

- Analog 1 (*SISSHSHFSSQSKSLK*): +20.3% docking score vs. lead
- Analog 16 (*AISAHSHFSSQSKSLP*): cleavage sites reduced 9 $\rightarrow$ 4

Full pipeline details available at github.com/kaih-b/peptide-analogs.

Peptide Generation Workflow



Unnatural Amino Acid Substitution

Five analogs were designed from the top candidates by substituting unnatural amino acids at potential protease-labile positions while preserving CD109 anchors. All are N-terminal acetylated to improve stability.

Unnatural Amino Acid	Replaces	Rationale
Pentafluorophenylalanine	Phe9 anchor	Maintains π -stacking anchor; blocks chymotrypsin/pepsin
Ornithine / D-Lysine	Lys	Retains +1 charge; resists trypsin
α -Methylarginine	Arg	Retains +1 charge; α -methyl blocks cleavage; stabilizes helix
L-tert-leucine	Leu	tert-butyl blocks chymotrypsin/pepsin; stabilizes helix
D-Norleucine	Met17	Blocks carboxypeptidase; removes Met oxidation liability
D-Serine	Ser1	Blocks N-terminal aminopeptidase

Cationic positions use three substitutions (Orn, D-Lys, α -Me-Arg). All improve protease resilience while sampling effects on binding.

Automated Solid-Phase Peptide Synthesis

Peptides were synthesized on rink amide resin (C-terminal amidated) via automated SPPS in CEM Liberty Blue (left), cleaved and deprotected in CEM Razor (right), precipitated in cold diethyl ether, and isolated via centrifugation. Crude product confirmed by LC-MS and purified by NCATS analytical team.



Structural Peptide Docking

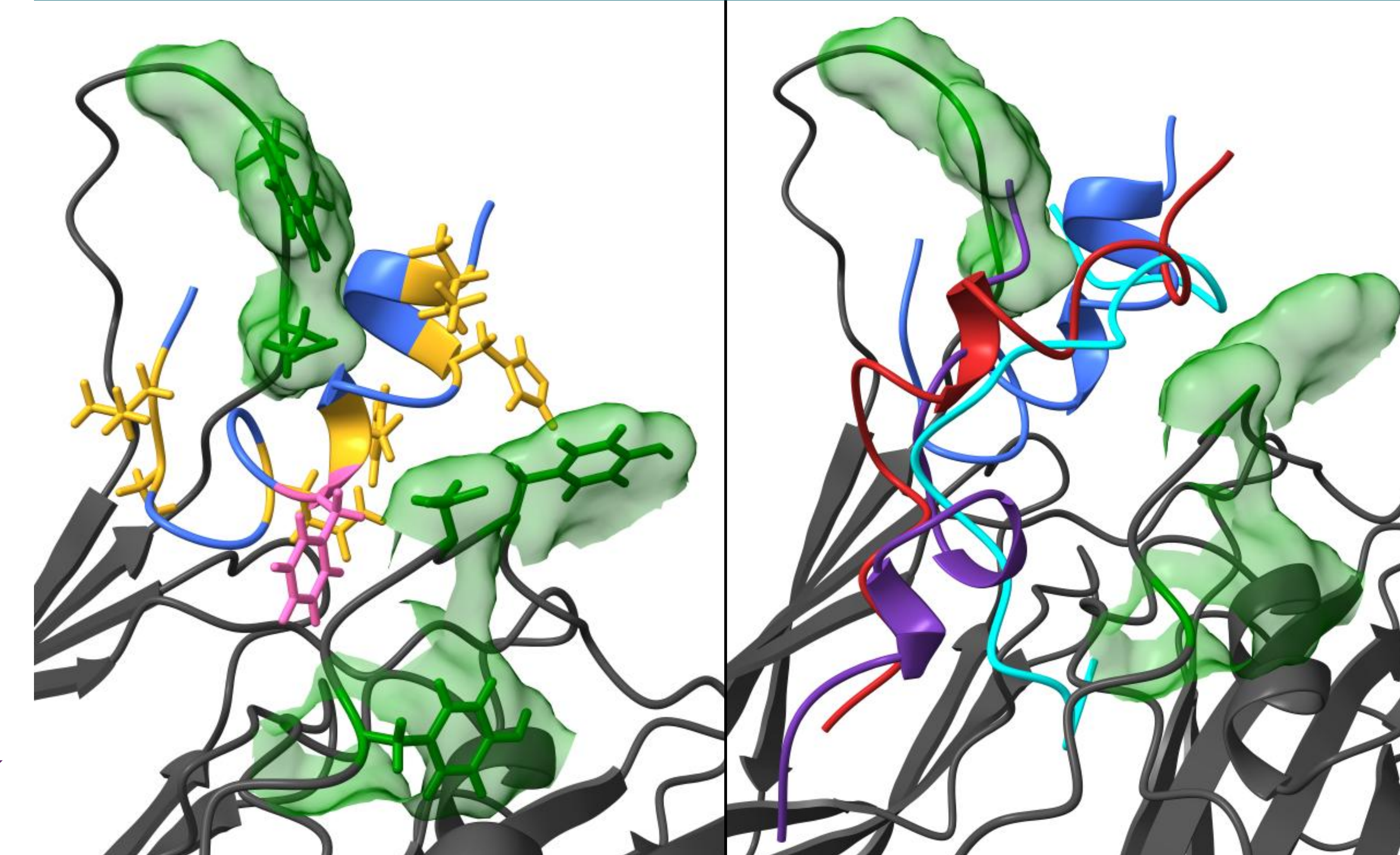


Figure 1: Lead peptide docked to target CD109 surface. Frozen peptide residues and Phe9 peptidase liability highlighted.

Figure 2: Analog 1, Analog 14, Analog 25 overlaid on lead peptide and docked to CD109. Representative poses selected.

Synthesized Analog Sequences

Natural:

SISSHSHFSSQSKSLK-[NH ₂]	LIAAHHAHFHKQSKSLP-[NH ₂]
LIGSHSHFHSQSRSL-[NH ₂]	AISAHSHFSSQSKSLP-[NH ₂]
SIASHAHFKHQAKSLM-[NH ₂]	IISHHSHFHKQALSLP-[NH ₂]
SISSHSHFQKQALSLP-[NH ₂]	LIAHSHSHFQKQAKSL-[NH ₂]

Unnatural Amino Acid Substitution:

[Ac]-(D-Ser)-ISSHSH-(F ₅ -Phe)-S-(D-Lys)-QA-(tLeu)-SLP-[NH ₂]
[Ac]-AISAHSSH-(F ₅ -Phe)-SSQS-(Orn)-SLP-[NH ₂]
[Ac]-IISHHSH-(F ₅ -Phe)-(D-Lys)-(Orn)-QALS-(tLeu)-P-[NH ₂]
[Ac]-LIGSHSHH-(F ₅ -Phe)-HSQS-(α -Me-Arg)-S-(tLeu)-(tLeu)-[NH ₂]
[Ac]-SIASHAH-(F ₅ -Phe)-(Orn)-HQA-(D-Lys)-SL-(D-Nle)-[NH ₂]

Current & Future Studies

- Validate the optimization workflow and identify leads with improved potency, stability, and safety profiles
- Evaluate the synthesized peptides with *in vitro* ADME assays and select the most promising candidates for *in vivo* pharmacokinetic studies
- Evaluate CD109 binding and CD109-CD26 blockade in cell-based assays, and their ability to enhance Th1/Th17 effector responses
- Advance the best-performing leads into *in vivo* efficacy studies

References & Acknowledgements

- N. Merle *et al.*, Complement-related CD109 averts autoimmunity but limits tumor control through Th1 and Th17 restraint. *Manuscript submitted*.
- J. Dauparas *et al.*, Robust deep learning-based protein sequence design using ProteinMPNN. *Science* **378**, 49-56 (2022). DOI:10.1126/science.add2187

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